



Ops-Management-Dashboard — Cheat Sheet

Chain C · Project C-3 · dense one-page recap — pairs with docs/interactive/index.html

1 · HTTP vs WebSocket

Model	Who initiates
HTTP polling	Client always asks
WebSocket	Either side, anytime, after one handshake

2 · ConnectionManager

- `active_connections: dict[int, tuple[WebSocket, str]]` — keyed by `user_id`
- `broadcast(event)` filters by `event.target_role`
- **Test trap:** proving "nothing received" with a timeout the test client doesn't support silently always passes — prove ordering instead

3 · RBAC

Caller role	POST /tickets/{id}/assign
STAFF	403 Forbidden
MANAGER	200 OK

- `require_manager` = a dependency, not an inline `if` — runs before the route body

4 · Real-Time UI

- Event arrives → `invalidateQueries(["tickets"])` → refetch → re-render
- Match by `event_type` `PREFIX ( ticket_*)`, not exact type — new event types just work
- `closedByEffectCleanup` flag stops an unmount from starting an immortal reconnect loop

Event protocol

Field	Meaning
<code>event_type</code>	e.g. "ticket_created", "ticket_assigned"
<code>payload</code>	the changed object
<code>actor_id</code>	who did it
<code>target_role</code>	"MANAGER" / "STAFF" / "ALL" — who should receive it